

Curriculum Vitae
Ho Man Colman Leung

colman.leung@columbia.edu
colmanleung.com  [LinkedIn](#)  [Google Scholar](#)

Research Interests My research focuses on physiological sensing for health and human-computer interaction applications. Guided by physical principles and natural phenomena, I carefully design sensors to capture high-quality signals and develop computational pipelines to extract meaningful biomarkers. Through hardware–software co-design, I integrate novel sensors, ubiquitous computing, and aesthetics into robust and reliable systems.

Education

PhD in Computer Science 2022 - Present
Columbia University | United States
Dartmouth College | United States 2020 - 2022
Advisor: [Prof. Xia ZHOU](#)

MPhil in Computer Science and Engineering 2015 - 2018
The Chinese University of Hong Kong | Hong Kong
Supervisors: [Prof. Chi-Wing FU](#) and [Prof. Pheng-Ann HENG](#)
Thesis: Cross-Device Authentication via Motion Co-analysis with a Smartwatch in a Multi-user Multi-device Environment

BSc in Computer Science 2009 - 2012
The Chinese University of Hong Kong | Hong Kong

Selected Publications

Clinical Evaluation of a Polarization-based Optical Noninvasive Glucose Sensing System
[Ho Man Colman LEUNG](#), Chengyue Gong, Luke Geiser, Emily E Fivekiller, Nam Bui, Tam Vu, Temiloluwa Prioleau, Gregory P Forlenza, Qiang Liu, Xia Zhou
Scientific Reports 15.1 (2025): 8877. [\[PDF\]](#)

My Pillow Knows My Sleep: Sleep Monitoring with Computational Fabrics in the Pillowcase
Qijia Shao, Junxiao Chen, [Ho Man Colman Leung](#), Meiqi Zhao, Ruoyu Xu, Jiting Liu, Lisa Maria DiSalvo García, Xiaofan Jiang, Marie-Pierre St-Onge, Xia Zhou
Proceedings of the ACM on IMWUT 9.3 (2025): 1-26. [\[PDF\]](#)

Fabric Sensing of Intrinsic Hand Muscle Activity
Katelyn Lee, Runsheng Wang, Ava Chen, Lauren Winterbottom, [Ho Man Colman Leung](#), Lisa Maria DiSalvo, Iris Xu, Jingxi Xu, Dawn M Nilsen, Joel Stein, Xia Zhou, Matei Ciocarlie
2025 International Conference On Rehabilitation Robotics (ICORR). IEEE, 2025. [\[PDF\]](#)

Joey: Supporting Kangaroo Mother Care with Computational Fabrics
Qijia Shao, Jiting Liu, Emily Bejerano, [Ho Man Colman Leung](#), Jingping Nie, Xiaofan Jiang, Xia Zhou
Proceedings of the 22nd Annual International Conference on Mobile Systems, Applications and Services. 2024. [\[PDF\]](#)

Noninvasive Glucose Sensing in Vivo
[Ho Man Colman LEUNG](#), Gregory P Forlenza, Temiloluwa O Prioleau, Xia Zhou
Sensors 23.16 (2023): 7057. [\[PDF\]](#)

Work Experience	Research Associate 2018 - 2020 <i>Department of Computer Science and Engineering, CUHK Hong Kong</i> Developed a robust multi-camera, multi-object 6DoF real-time tracking system for medical simulators, with performance boosted through CUDA parallelization and pipelined execution.
	Software Engineer 2012 - 2015 <i>Dracaena Life Technologies Co., Ltd. Hong Kong</i> - Designed an anatomy visualization tool with stereoscopic 3D and Leap Motion mid-air gestures for immersive learning; deployed at PolyU School of Nursing with positive user study results. - Developed a Traditional Chinese Acupuncture Simulation System by extending an Interventional Sonography Trainer, adding rotary encoders and Arduino-based sensors to capture needle rotation. - Implemented stereoscopic rendering and head tracking 3D simulation using Kinect.
Service & Outreach	Organizing Member & Technical Instructor You(th)Fab 2026 - Present - Co-organizing a NYSCA-funded initiative teaching digital fabrication to high schoolers. - Instructing students in CAD for 3D printing and laser cutting.
	Volunteer Mentor Columbia PhD Application Support (PAR) 2023-2025 Providing feedback on personal statements and CVs for prospective PhD applicants to Columbia University.
	Technical Reviewer ACM IMWUT, ACM MobiSys
Teaching	CSEE4119 Computer Networks - Teaching Assistant Columbia University Spring 2024 & Spring 2025 CS60 Computer Networks - Teaching Assistant Dartmouth College Spring 2021 Final Year Project Tutor CUHK Fall 2016 - Spring 2017 CSCI3260 Principle of Computer Graphics - Teaching Assistant CUHK Spring 2016 Awarded the Certificate of Merit
	CSCI1130 Introduction to Computing using Java - Teaching Assistant CUHK Fall 2015
Skills	Software Development Programming: Python, C/C++, C#, Java, Objective-C, MATLAB, GLSL Libraries: PyTorch, TensorFlow, OpenCV, OpenGL, CUDA, Boost Platforms: iOS, Android, Arduino, Unity, Windows, Web
	Hardware Fabrication Design Tools: Autodesk Fusion, SolidWorks, EasyEDA, Inkscape Techniques: Rigid & Flex PCB Layout, SMD Soldering, 3D Printing, E-Textiles/Embroidery
	Designing Adobe Creative Suite (Photoshop, Illustrator, InDesign, Premiere Pro, After Effects)
	Spoken Languages Cantonese, English, Mandarin, Spanish (Beginner)
Interests	Dragon Boat, Badminton, Piano, Cello, Graphic Design